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Image quality evaluation

Digital Forensic

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1 Introduction

Human eyes can't detect if an image is different from another. But in some cases, it can be critical to know if the image is different from another, especially in compression domain where the goal is to conserve a good quality of image with a maximum compression. In this work, we will work on the image quality evaluation using several methods.

2 Methodology

To make image quality evaluation, we will use the MSE, PSNR, SSIM and LPIPS methods between two images, original image and modified image.

2.1 MSE (Mean Square Error)

This is a simple image quality evaluation that basicaly compute the mean between the two images and take the square of it. Mathematicaly, it can be represented by formula 1 for two images I and K of size $m \times n$.

$$\text{MSE} = \frac{1}{mn} \sum_{i=1}^{m} \sum_{j=1}^{n} [I(i,j) - K(i,j)]^2 \tag{1}$$

It give the sum of the square mean of each pixel, so a little value mean that the two images are very similar.

2.2 PSNR

The Peak Signal Noise Ratio (PSNR) is a measure that evaluate the quality of reconstructed images for instance after compression by comparing the maximal strength of the signal to the strength of the noise generated during reconstruction that affect the image.

Mathematicaly, it can be represented by formula 2 for two images I and K of size $m \times n$.

$$\text{PSNR} = 10 \cdot \log_{10} \left(\frac{MAX_I^2}{\text{MSE}} \right) \tag{2}$$

Where MAX_I^2 is the maximum value of a pixel.

As said in section 2.1, a little value of the MSE indicate that the two images are very similar. So as we have $\frac{MAX_I^2}{\text{MSE}}$, it means that when images are very similar, PSNR is high.

2.3 SSIM

The Structural Similarity Index (SSIM) is a perceptual metric used to quantify the similarity between two images, focusing on human vision. It measure the perceptual differ-

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ences between two images, based on visible structure of images. So it's not a pixel-wise method, but more a method useful in term of visual perception of human.

Mathematicaly, it can be represented by formula 3 for two images I and K of size $m \times n$.

$$\mathrm{SSIM}(I, K) = \frac{(2\mu_I\mu_K + C_1)(2\sigma_{IK} + C_2)}{(\mu_I^2 + \mu_K^2 + C_1)(\sigma_I^2 + \sigma_K^2 + C_2)} \tag{3}$$

With:

- μ_I the mean of image I
- μ_K the mean of image K
- σ_I^2 the variance of image I
- σ_K^2 the variance of image K
- σ_{IK} the covariance between image I and image K
- C_1, C_2 constant values to stabilize the division

If we suppose that image I is image K, the SSIM give us:

$$\mathrm{SSIM}(I, I) = \frac{(2\mu_I\mu_I + C_1)(2\sigma_I^2 + C_2)}{(\mu_I^2 + \mu_I^2 + C_1)(\sigma_I^2 + \sigma_I^2 + C_2)} = \frac{(2\mu_I^2 + C_1)(2\sigma_I^2 + C_2)}{(2\mu_I^2 + C_1)(2\sigma_I^2 + C_2)} = 1$$

So a value closest to 1 indicate that the two images are perceptualy very similar.

3 Implementation

For the implementation, I use `python` with OpenCV to load image and to make the jpeg compression.

Else, `numpy` library is used to make operations on image such as adding a gaussian noise, compute the MSE (ref section 2.1) or the PSNR (ref section 2.2).

In my code, I work only with images in range [0, 255].

4 Results

To achieve my results, I have downloaded photos on internet and I have tried to not have digital image or generated image. On generated images or on digital images, perhaps the quality measures will be different due to another structure of our image, with no natural noise etc...

The results are presented below in increasing order: first image with size 256x256, second with size 512x512, ...

Original image

```
==============================================
Gaussian noise with sigma = 1
MSE: 1.3290252685546875
PSNR: 46.89547122671948
SSIM: 0.996671598787309

==============================================
Gaussian noise with sigma = 5
MSE: 24.632965087890625
PSNR: 34.21563669540209
SSIM: 0.9368046645688801

==============================================
Jpeg compression 90
MSE: 7.6191864013671875
PSNR: 39.31171762266033
SSIM: 0.9900354972307785

==============================================
Jpeg compression 50
MSE: 50.512420654296875
PSNR: 31.096821796132584
SSIM: 0.879297637296279
```

Original image

```
==============================================
Gaussian noise with sigma = 1
MSE: 1.3299369812011719
PSNR: 46.89249298370755
SSIM: 0.9911128397733764

==============================================
Gaussian noise with sigma = 5
MSE: 24.750934600830078
PSNR: 34.19488758231862
SSIM: 0.8398336960936418

==============================================
Jpeg compression 90
MSE: 4.24200439453125
PSNR: 41.85509246777023
SSIM: 0.9729234131095675

==============================================
Jpeg compression 50
MSE: 12.729198455810547
PSNR: 37.08279303418355
SSIM: 0.9368902127255937
```

Original image

```
================================================
Gaussian noise with sigma = 1
MSE: 1.308967
PSNR: 46.96151663055549
SSIM: 0.9992328709453819

================================================
Gaussian noise with sigma = 5
MSE: 24.183222
PSNR: 34.29566198189295
SSIM: 0.9847707616027714

================================================
Jpeg compression 90
MSE: 7.673377
PSNR: 39.28093824859601
SSIM: 0.9962652145352185

================================================
Jpeg compression 50
MSE: 69.434526
PSNR: 29.715048857631558
SSIM: 0.9234532550599298
```


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Original image

```
==============================================
Gaussian noise with sigma = 1
MSE: 1.333908
PSNR: 46.879544836638125
SSIM: 0.9878534586886185

==============================================
Gaussian noise with sigma = 5
MSE: 24.8138495
PSNR: 34.183862170341285
SSIM: 0.7917822684012716

==============================================
Jpeg compression 90
MSE: 0.223223
PSNR: 54.64341420340431
SSIM: 0.9983469290136069

==============================================
Jpeg compression 50
MSE: 2.58476375
PSNR: 44.00659506575346
SSIM: 0.9904252758128195
```


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Original image

```
================================================
Gaussian noise with sigma = 1
MSE: 1.3254852294921875
PSNR: 46.907054683203214
SSIM: 0.9966655078835887

================================================
Gaussian noise with sigma = 5
MSE: 24.794570922851562
PSNR: 34.18723763766099
SSIM: 0.9363550521306953

================================================
Jpeg compression 90
MSE: 7.6191864013671875
PSNR: 39.31171762266033
SSIM: 0.9900354972307785

================================================
Jpeg compression 50
MSE: 50.512420654296875
PSNR: 31.096821796132584
SSIM: 0.879297637296279
```

Original image

```
==============================================
Gaussian noise with sigma = 1
MSE: 1.3339157104492188
PSNR: 46.87951973298876
SSIM: 0.9911362201044676

==============================================
Gaussian noise with sigma = 5
MSE: 24.811599731445312
PSNR: 34.184255944940666
SSIM: 0.839747761803631

==============================================
Jpeg compression 90
MSE: 4.24200439453125
PSNR: 41.85509246777023
SSIM: 0.9729234131095675

==============================================
Jpeg compression 50
MSE: 12.729198455810547
PSNR: 37.08279303418355
SSIM: 0.9368902127255937
```

Original image

```
==============================================
Gaussian noise with sigma = 1
MSE: 1.305721
PSNR: 46.97229971911118
SSIM: 0.999236040990072

==============================================
Gaussian noise with sigma = 5
MSE: 24.120372
PSNR: 34.30696359377349
SSIM: 0.984819031329949

==============================================
Jpeg compression 90
MSE: 7.673377
PSNR: 39.28093824859601
SSIM: 0.9962652145352185

==============================================
Jpeg compression 50
MSE: 69.434526
PSNR: 29.715048857631558
SSIM: 0.9234532550599298
```


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Original image

```
================================================
Gaussian noise with sigma = 1
MSE: 1.33406375
PSNR: 46.87903777455523
SSIM: 0.9878587639994199

================================================
Gaussian noise with sigma = 5
MSE: 24.806886
PSNR: 34.18508110013509
SSIM: 0.7917118713149075

================================================
Jpeg compression 90
MSE: 0.223223
PSNR: 54.64341420340431
SSIM: 0.9983469290136069

================================================
Jpeg compression 50
MSE: 2.58476375
PSNR: 44.00659506575346
SSIM: 0.9904252758128195
```


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5 Discussion

As mentionned in part 2:

- if MSE is low, the two images are very similar
- if PSNR is high, the two images are very similar
- if SSIM is close to 1, the two images seems to be similar

So we can constate that when we increase the noise, the MSE increase, the PSNR decrease and the SSIM go far from 1. It's the same for JPEG compression...

For the PSNR, as it is defined as a ratio between max possible intensity of a pixel and the MSE, and as the maximum intensity possible for all pixel is always the same for all images because they are all in range from 0 to 255, it means that the PSNR give us basically an information similar to the MSE, but more "human readable". So we look at the MSE and for the PSNR, it's the same results (but be carefull, it goes in inverse order: low is bad and high is good !).

If we compare the results side by side, we can constate that the MSE is always the same for images of all sizes when adding noise. I think it's basically because we finally measure the mean of the gaussian noise added and this noise is always generated with same gaussian distributions.

We can also see that for JPEG compression, the MSE is, for instance for Lena image, closest between compression 90 and compression 50 whereas for the forest image, the MSE have a huge difference. I think it's due to the structure of the image. In case of Lena image, we have a lot of smooth part in our image whereas with the forest, we don't have any smooth part: we have alway variations due to leafs of trees. So the compression loose data when image is complex with a lot of details, like a forest.

If we look at the SSIM between images, we can see that a great variation of the MSE doesn't mean a great variation of the SSIM. For instance, if we look at the forest image, more particulary to the JPEG compression, there is a huge difference between the MSE, but the SSIM is very close because it's only 0.06 of difference between the two results obtained. It's perhaps due to the compression: the goal of the compression is to reduce the size of the image in memory by conservating a good similarity with the original image in an "human vision" point of view. And here we can say that the goal is reached because the compression have kept perceptible similarities because the SSIM is close to 1 and since the SSIM measure the perceptible difference, it is not too impacted by the compression, contrary to the MSE.

Another observation that we can make is that the SSIM is more impacted by the gaussian noise when the image is smoothy because when we look at the forest image, when we add the noise, the SSIM changes only from 0.01 whereas with the Lena or the deer image, which are more smoothy, when we add the noise, the SSIM changes about approximatively 0.2 which is a big difference. It's due to the fact that noise is more visible on smoothy images than on complex images. And as SSIM is based on perceptible differences, this explain the difference observed.

6 Conclusion

The PSNR and the MSE give same informations but not in the same way. PSNR is designed to be more human readable than MSE. It's a pixel-wise difference whereas the SSIM method give a perceptible difference, based on human vision.

In the analysis of the images of different sizes, we have seen that adding gaussian noise give always similar values for MSE and PSNR and for images of different sizes. It's due to the analysis of the gaussian noise, which follows always the same distribution, by the two method. We have also seen that when an image is complex, making a JPEG compression will cause a loss of informations but not visible ones. Finally, we have seen that on smoothy images, adding a noise is more perceptible and give more difference between the SSIM obtained.

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